



PRODUCT NAME

Industrial-grade Dry Kitchen Cleaner Powder Formula

Description

A liquid solution designed for deep cleaning in kitchen environments with effective degreasing capabilities. This non-abrasive formula removes tough stains swiftly and is packaged in an eco-friendly manner.

Technical Specifications

pH Level:	7
Viscosity:	Low
Shelf Life:	24 months
Batch Size:	50-200 liters
Processing Time:	2 hours
Temperature:	60°C during initial phase
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Mixing tank, stirrer, homogenizer, temperature control system
Certification:	EcoCert, ISO 9001

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	78.0%	Solvent
2	Sodium Lauryl Sulfate	Sodium Lauryl Sulfate	10.0%	Surfactant for degreasing
3	Sodium Citrate	Sodium Citrate	4.0%	pH buffer
4	Cocamidopropyl Betaine	Cocamidopropyl Betaine	3.0%	Foaming agent
5	Sodium Carbonate	Sodium Carbonate	2.5%	Alkalinity agent for stain removal
6	Sodium Metasilicate	Sodium Metasilicate	1.5%	Enhance cleaning performance
7	Limonene	Limonene	0.5%	Natural fragrance
8	Citric Acid	Citric Acid	0.5%	pH adjuster
Total			100%	

Manufacturing Process

Phase 1: Water Phase

- Heat Water (Aqua) to 60°C in a mixing tank.
- Slowly add Sodium Lauryl Sulfate while stirring until fully dissolved.

Phase 2: Active Phase

- Add Sodium Citrate and mix until fully dispersed.
- Blend in Cocamidopropyl Betaine and continue mixing.

Phase 3: Additives Phase

1. Introduce Sodium Carbonate and Sodium Metasilicate to the blend, ensuring even dispersion.
2. Add Limonene for fragrance.
3. Use Citric Acid to adjust pH to exactly 7.0.

Phase 4: Cooling Phase

1. Cool the mixture to room temperature.
2. Homogenize the solution to ensure uniformity.

Required Equipment

Mixing tank, stirrer, homogenizer, temperature control system

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: EcoCert, ISO 9001

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 50-200 liters.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply directly on kitchen surfaces, wait for a few seconds, and wipe away with a clean cloth. Rinse with water if necessary.